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A Physics Inspired Data-Driven Model for Candidate Selection and Performance Assessment of Refracturing in Unconventional Wells

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Abstract

This study presents a physics-inspired, data-driven framework to improve refracturing candidate selection and assess post-job performance in unconventional reservoirs. By integrating advanced reservoir and well diagnostics with data-driven techniques, the proposed approach enhances the ability to identify underperforming wells with high remaining potential. A key innovation is the introduction of the Refrac Well Potential (RWP) factor, which combines Reservoir Potential (RP) and Completion Quality (CQ) to rank candidate wells. RP is derived from dynamic drainage volume (DDV), reservoir flow regimes, well spacing, average reservoir pressure, and injected fluid volumes, while CQ is based on productivity index decline trends and cumulative production. To evaluate complex production behavior and refracturing impact, advanced modeling techniques such as Sparse Identification of Nonlinear Dynamics (SINDy) and multi-segmentation analysis are applied. Field validation shows strong alignment between pre-job RWP rankings and post-job production outcomes, with high-RWP wells consistently delivering greater incremental production. Additionally, the framework enables performance comparisons across different refracturing strategies, supporting more capital-efficient decisions. This methodology offers a practical and validated workflow to optimize refracturing investments and maximize recovery in unconventional plays.

Introduction

Unconventional plays in the US have undergone extensive development through horizontal drilling and multi-stage hydraulic fracturing. While initial production rates are often high, wells typically experience steep declines as fractures degrade, depletion advances, and parent–child interactions emerge. Refracturing—re-stimulating a previously completed well—offers a cost-effective means to restore productivity or access bypassed rock without drilling new wells. Low-cost restimulation methods, such as acidizing, aim to remove near-wellbore damage and recover lost productivity. However, outcomes vary widely, and economics are highly sensitive to both candidate selection and completion design. Recent basin-scale analyses have quantified how parent–child spacing and proppant intensity influence frac-hit risk and post-job uplift (He et al., 2023; Parekh et al., 2025). These patterns highlight the need for workflows that (i) diagnose and quantify effects on dynamic drainage volume and reservoir pressure, (ii) segment pre-/post-

event behavior, and (iii) connect these diagnostics to a ranking metric that guides refrac and restimulation decisions at both well and pad scale.

Hybrid, physics-guided Transient Well Performance (TWP) method—combining diffusive time-of-flight (DTOF), sequential pseudo-steady/material balance concepts, and transient productivity index analysis—is used to estimate evolving drainage volume, average reservoir pressure, and well deliverability from standard production and pressure data (Sankaran et al., 2022; Mohamed & King, 2024;). While TWP has shown strong agreement with reliable BHP estimates, its accurate application requires careful handling of rate normalization and segmentation, especially when rates and pressures fluctuate significantly. To further sharpen event detection and flow regime classification, the Sparse Identification of Nonlinear Dynamics (SINDy) framework has emerged as a powerful data-driven approach for deriving governing dynamic equations from time-series data (Brunton et al., 2016; Abdullah & Christofides, 2023). By tracking shifts in rate-normalized pressure (RNP) versus material balance time (MBT) slopes, SINDy can capture subtle transitions between bilinear, linear, and sub-linear (pseudo-steady) regimes, while advanced variants such as abrupt-SINDy enable rapid re-identification of governing dynamics following sudden changes—making them particularly effective for detecting refrac events, frac-hits, and other transient reservoir responses (Quade et al., 2018; Corbetta, 2020).

Early refracturing candidate selection frameworks primarily relied on production-based heuristics and rate-transient analysis (RTA) plots to identify underperforming wells, often using decline curve deviations or type curve mismatches as proxies for remaining potential (Roussel & Sharma, 2013). Subsequent methods incorporated multi-criteria decision-making (MCDM) tools such as the Analytic Hierarchy Process (AHP), grey relational analysis, and fuzzy comprehensive evaluation to integrate geological, completion, and production indicators into composite ranking scores, which generally outperform ad hoc selection and have been validated in multiple tight oil and shale gas field studies (Guo et al., 2019; Xiong et al., 2022). More recently, machine learning models trained on historical refracturing outcomes across multiple basins have been introduced to improve statistical robustness (Velasquez & Tabatabaie, 2023). However, most existing approaches still lack (i) dynamic drainage diagnostics to quantify changes in effective stimulated volume, (ii) automated event detection to distinguish refracs from interference events such as frac-hits, and (iii) formation-specific calibration to address geological heterogeneity and most importantly (iv) robust and automated way to rank candidates for refrac. Meanwhile, interference risk, well spacing, and completion intensity are increasingly recognized as critical determinants of refracturing success; in tightly spaced unconventional developments, pressure depletion and stress redistribution in parent wells can alter fracture propagation paths, increasing frac-hit likelihood and causing asymmetric fracture growth toward depleted zones (Maqbool et al., 2022). Targeted refracturing of a parent well ("protection refrac") can recharge reservoir pressure and redistribute stress fields, thereby reducing interference risk and, in some cases, enhancing child well performance—highlighting the need for refrac ranking systems that integrate flow regime, reservoir pressure, spacing, depletion dynamics, and prior completion history to fully assess both opportunity and risk.

Despite these advances, existing refracturing candidate selection frameworks still face significant limitations. Many MCDM and machine learning approaches treat geological, completion, and production attributes purely as statistical predictors, without explicitly linking them to underlying reservoir physics such as drainage volume evolution, pressure state, and flow regime transitions. This lack of physical grounding can reduce interpretability and lead to inconsistent predictions across formations. Furthermore, traditional decline curve and RTA methods often struggle to distinguish refracturing from other events such as frac-hits or operational shut-ins, resulting in event misclassification. Most workflows also focus narrowly on single-well selection, overlooking pad-scale optimization, which is crucial for mitigating interference risk, improving operational efficiency, and maximizing economic returns.

To address these challenges, this study develops and validates a physics-inspired, data-driven refrac candidate ranking and assessment workflow. The approach integrates transient well performance (TWP)

analysis to quantify evolving reservoir pressure and depletion trends with multi-segment dynamic drainage volume (MS-DDV) diagnostics, enhanced by sparse identification of nonlinear dynamics (SINDy) for automated event detection and flow regime classification. A newly defined Refrac Well Potential (RWP) metric combines dynamic reservoir diagnostics with static spatial and completion parameters, enabling ranking at both well and pad scales. The methodology is applied to a major US onshore unconventional play and validated against post-job outcomes from refracturing treatments. Results demonstrate that integrating physics-based diagnostics with data-driven ranking can significantly improve candidate selection accuracy, enhance operational decision-making, and provide a scalable framework for unconventional reservoir development.

Methodology

Overview

The methodology integrates physics-based reservoir diagnostics with data-driven modeling to systematically screen refracturing candidates and evaluate their performance. The core of the workflow is the Refrac Well Potential (RWP) metric, which incorporates both static and dynamic characteristics of the reservoir and completion to rank wells for refracturing suitability. The process includes multi-segmented dynamic drainage volume (MS-DDV) analysis, reservoir pressure modeling, productivity assessment, and post-job validation.

As shown in Figure 1, the refracturing well candidate selection and evaluation procedure involves three steps. Firstly, transient well performance (TWP) is applied by detecting and classifying well events and its effect on well performance by capturing dynamic drainage volume, average reservoir pressures and productivity index estimates. Following this, the Refrac/Restim Candidate Screening phase involves selecting a group of wells based on attributes such as user defined formation, pad, and cumulative production. Cluster analysis is used to evaluate trends and heterogeneity, after which Refrac Well Potential (RWP) is calculated for each candidate and ranked accordingly. Finally, the RWP scores and rankings are validated using post-refrac performance data from refrac wells, providing feedback for refining the candidate selection criteria. Each step is described in detail in the following sections.

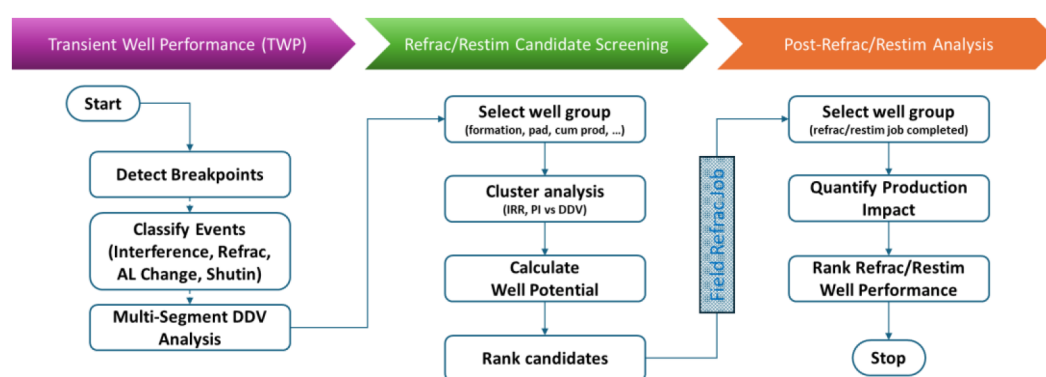


Figure 1—Refrac candidate selection and post-job analysis workflow

Transient Well Performance (TWP)

In our previous studies, we developed a comprehensive hybrid reservoir modeling framework for the analysis of transient well performance (TWP). This framework is founded on a novel formulation that integrates diffusive time-of-flight calculations with sequential pseudo-steady-state (SPSS) material balance principles and transient productivity index (PI) concepts, thereby enabling rigorous production diagnostics and forecasting (Molinari & Sankaran, 2021, Zalavadia et al. 2025). The methodology has been extensively

validated through application to thousands of wells across multiple shale plays and tight reservoirs in North America.

Hybrid Reservoir Model. The fundamental premise of TWP analysis is that the drainage volume expands over time, albeit with an indeterminate geometry. The diffusive time-of-flight method is employed to estimate this evolving drainage volume while enforcing a no-flow outer boundary, with cumulative production data used to account for reservoir withdrawals. Under the SPSS assumption, high-resolution material balance calculations (e.g., daily timesteps) provide a dynamic characterization of reservoir pressure depletion and average pressure trends within the contacted drainage volume at any given time. This enables the determination of a transient PI that captures the true reservoir deliverability, serving as a critical input for pressure-aware, multiphase inflow performance relationship (IPR) calculations. The workflow is illustrated in Figure 2, which summarizes the key components: (1) estimation of dynamic drainage volume using diffusive time-of-flight, (2) computation of average reservoir pressure through a succession of pseudo-steady states, and (3) derivation of a transient, pressure-aware PI.

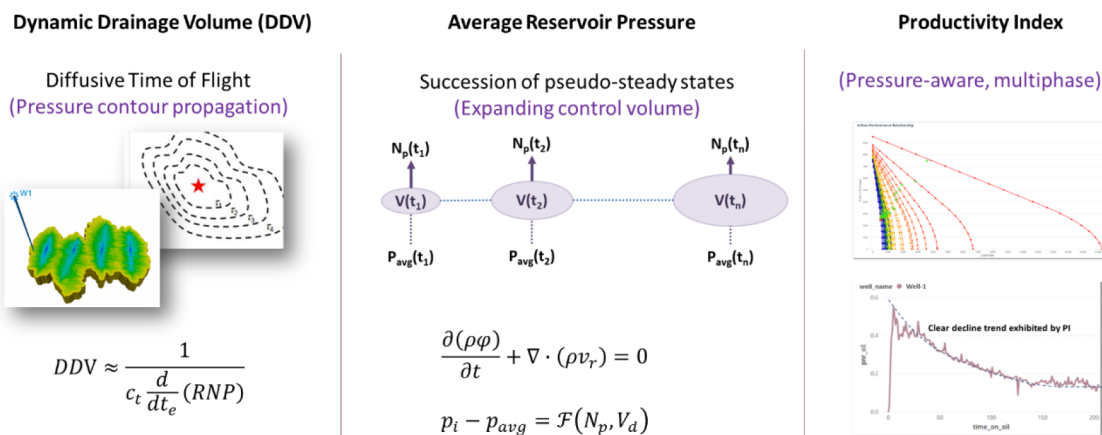


Figure 2—TWP workflow

Multi-Segmented Dynamic Drainage Volume (MS-DDV). Dynamic Drainage Volume (DDV) is a crucial parameter in Transient Well Performance (TWP) analysis, as it directly impacts the reservoir pressure and productivity calculations used for forecasting and production optimization workflows (Zalavadia et al. 2022). Until recently, the approach approximated drainage volume to monotonically increase with time which is true for wells that do not suffer from extraneous events such as frachits and refracs. In order to handle the more complex dynamics associated with frachits and refrac that fundamentally lead to an abrupt change in DDV of the well, an enhancement to the workflow was required that led to the development of MS-DDV.

An automated segmentation method was firstly developed based on Pressure-Normalized Rate (PNR) and water-cut trends (Zalavadia et al. 2025) to detect sections of the well history post frachit or refrac events. After dividing the production history into multiple segments, a more precise DDV calculation is performed in each segment by computing Rate Normalized Pressure (RNP) derivatives using Sparse Identification of Nonlinear Dynamics (SINDy) approach. The idea is to propose flow regime-based time features for expressing RNP inspired by RTA theory and SINDy can identify the most relevant features for individual wells with time. The fundamental idea behind SINDy is to use sparse regression techniques to identify the governing differential equations that describe the system's dynamics. Unlike traditional modeling approaches, which require predefined physical models, SINDy learns the system's behavior directly from data.

The model identifies the governing equations leading to distinct decline trends of different flow regimes, i.e., bilinear flow, linear flow, SRV/PSS flow, sub-linear or sub-radial flow etc. as shown in Figure 3. With the multi-segmented approach, DDV profile also shows an abrupt change at the onset of frachit or refrac that can help quantify the impact of the event on well's drainage as shown in Figure 3.

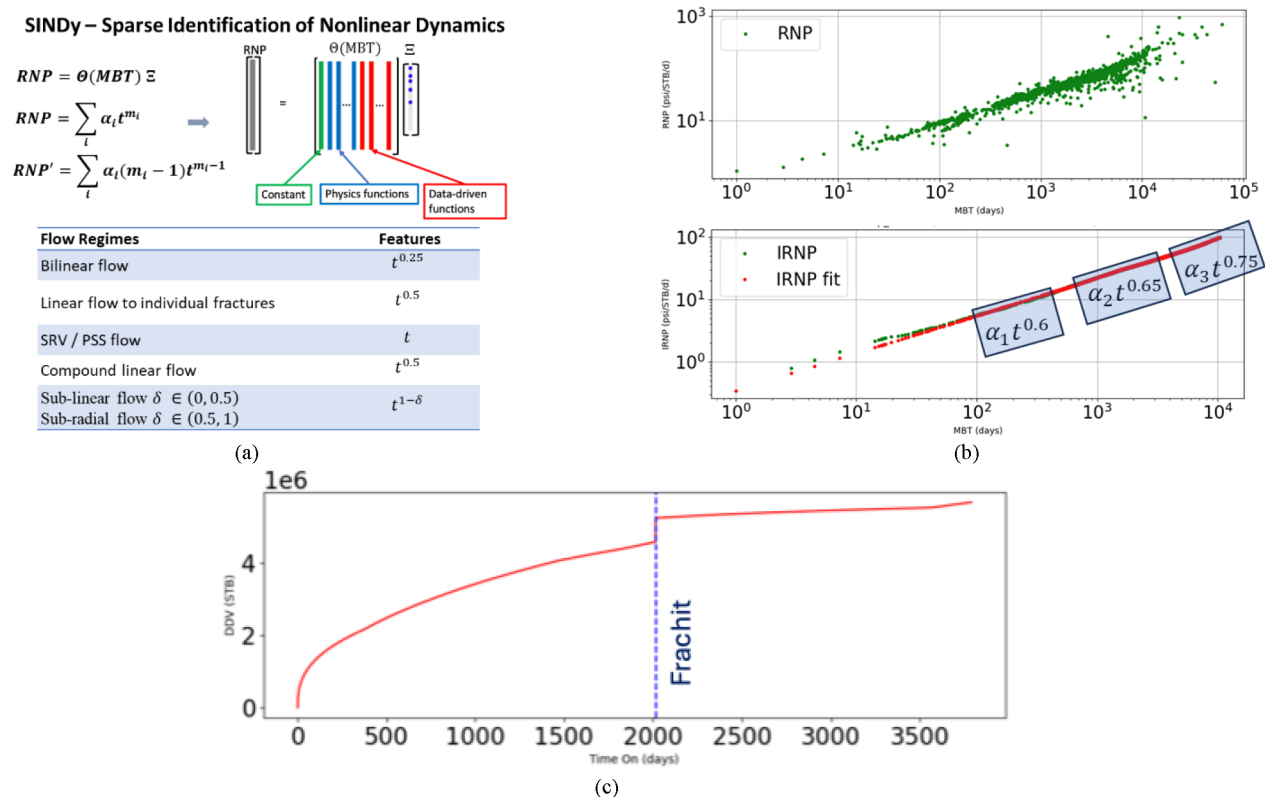


Figure 3—Multi-Segmented DDV modeling workflow

Refrac/Restim Candidate Screening

In evaluating candidates for refracturing, it is useful and practical to focus on wells with sufficient production history and indications of underperformance, as these provide both the necessary data for diagnostics and the greatest potential for improvement. A new ranking metric, the Refrac Well Potential (RWP), is introduced to systematically assess such wells. This approach integrates two primary factors: Reservoir Potential (RP) and Completion Quality (CQ). The underlying idea is to prioritize wells that exhibit high reservoir potential but relatively poor completion quality, since these are most likely to respond positively to refracturing. Conceptually, an ideal candidate would combine strong reservoir capacity—suggesting the ability to deliver higher production after intervention—with suboptimal completion characteristics, indicating room for improvement. The detailed formulation and methodology for this metric are described in the following sections.

Reservoir Potential (RP). Reservoir Potential (RP) reflects the potential of the reservoir to contribute towards the success of refracturing. An ideal refrac well candidate is expected to have higher DDV, higher well spacing, higher average reservoir pressure representing the energy of the reservoir. Larger fractures also contribute to higher DDV, thus, in order to represent the reservoir potential through DDV, injected fluid volume is used as a proxy to normalize the completion effects on DDV, as higher injected fluid volume would indicate larger fractures. Finally, reservoir potential is also influenced by fraction of producing time the well has been in linear or sublinear flow regimes or time it takes for the well to reach transitional flow regime that indicates fracture/well interference, indicating the higher that fraction the more likelihood

of untapped drainage volumes. These wells are good candidates for refracturing, as they may still have significant reserves in the reservoir that can be accessed with enhanced stimulation. An illustration of how this fraction to transitional flow is calculated is shown in Figure 4.

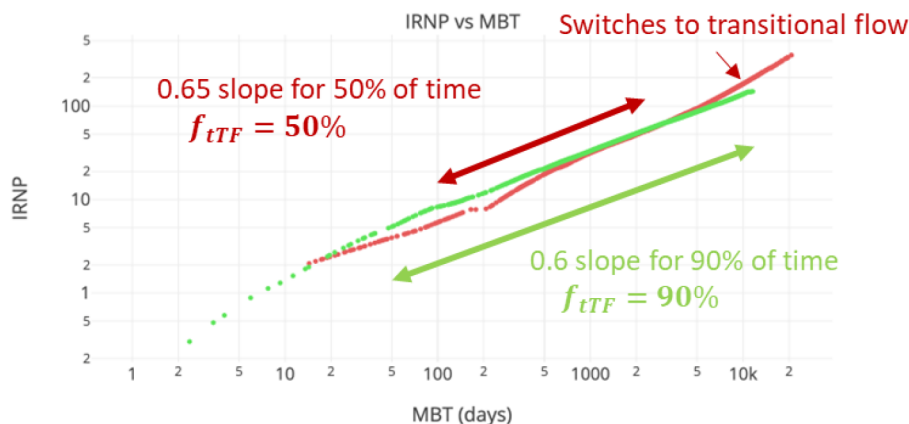


Figure 4—Fraction of producing time to reach transitional/BDF flow. 2 wells in red and green showing log-log IRNP vs MBT which uses the SINDy approach to estimate the dominant flow regimes with time.

As can be seen, green well has a slope of 0.6 for 90% of the production history and hence has $f_{tTF}=0.9$, indicating close to linear flow regime for majority of well life. The red well has a slope of 0.65 for 50% of the time before it switches to a transitional flow regime and hence has $f_{tTF}=0.5$. Then, RP can be calculated using Equation 1:

$$RP = \frac{DDV^{a_1}}{FV^{a_5}} \times f_{tTF}^{a_2} \times \bar{P}_{res}^{a_3} \times Spacing^{a_4} \times formation\ factor \quad \text{Equation 1}$$

where exponents a_1 to a_5 are unknowns that can be tuned based on analyzing the post-refrac performance as will be illustrated in the later section. If the post refrac performance is not available, setting them all equal to 1 is still found to be quite accurate to capture the reservoir potential. Table 1 shows the definition of RP parameters that need to be defined for refrac candidate wells. TWP approach illustrated above thus provides extremely useful insights to capture the reservoir potential for individual wells in the field.

Table 1—Definition of RP parameters

Parameters	Definitions
DDV	Dynamic Drainage Volume
Well Spacing	average effective spacing
$\bar{P}_{res}$	Average Reservoir Pressure
f_{tTF}	Fraction of producing time to reach transitional/BDF flow
FV	Injected Fluid Volume for normalization
$formation\ factor$	Weightage for formations to capture other properties that maybe important to characterize

Completion Quality (CQ). Completion Quality (CQ) refers to the prevalent effectiveness of completion in supporting productivity of the well. A poor or degraded completion quality negatively affects the well's ability to maintain high production rates over time, and thus, the well may benefit from refracturing or restimulation. The decline in Productivity Index (PI) during the initial production period is largely governed by the quality of completion job and near wellbore depletion, with poor completion quality and high depletion leading to rapid PI decline in the early well life (6 months-1 year), hence can be expressed as Equation 2. PI can be characterized using Arps hyperbolic decline as a function of cumulative liquids produced as has been shown in various other forecasting and production optimization applications (Molinari

and Sankaran, 2021; Sinha et al., 2023). Decline parameters are derived from early-time PI trends and subsequently used to compute a normalized PI value based on current cumulative liquid production. This approach isolates the effects of completion quality and depletion, avoiding reliance on raw PI values from later production stages, which may be influenced by additional factors such as multiphase flow effects or well interference. CQ can thus be calculated using Equation 3 below:

$$PI_{early-time} \propto f\left(\frac{CQ}{Depletion}\right) \Rightarrow CQ \propto f(PI_{early-time} * Depletion) \quad \text{Equation 2}$$

$$CQ = \left(\frac{PI_i}{(1 + bD_iL_p)^{\frac{1}{b}}} \right)^{b_1} * \left(1 - \frac{\bar{P}_{res}}{P_{init}} \right)^{b_2} \quad \text{Equation 3}$$

Similar as RP, exponents b_1 and b_2 are tuned based on post-refrac analysis of refrac wells shown later and if post-refrac analysis is not available, setting the exponents equal to 1 has been found to provide a reasonable approximation. Table 2 shows the definition of CQ parameters for a preferred refrac candidate well:

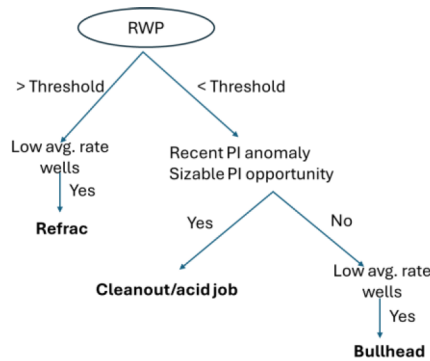
Table 2—Definition of CQ parameters

Parameters	Definition
PI_i	Initial PI from Arps fit on early time production
D_i	Arps PI decline rate on early time production
b	b factor from PI Arps fit on early time production, b
L_p	Total Cumulative Liquid Volumes for normalization
$\frac{\bar{P}_{res}}{P_{init}}$	Pressure depletion during early time production

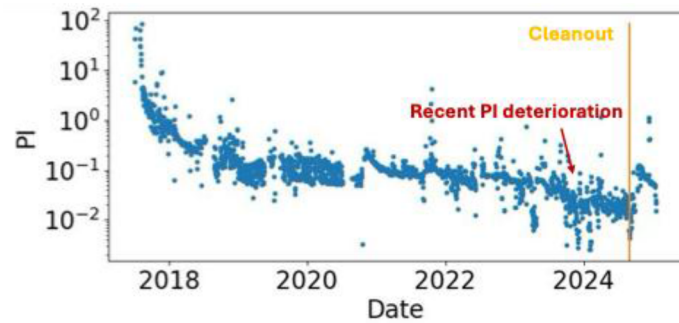
Refrac Well Potential (RWP). To determine the Refrac Well Potential (RWP), both RP and CQ are normalized based on the wells within the group under consideration. This normalization allows for a consistent ranking of wells within a group. Once RP and CQ are assessed and normalized, the RWP can be calculated to rank and prioritize the wells for refracturing. The wells with the highest RWP are those with high reservoir potential and poor completion quality, making them the best candidates for a refrac job. The normalization process takes into account group-specific factors, ensuring that wells are ranked based on their overall potential relative to others within the same group, leading to a more effective prioritization of refracturing candidates. RWP can be calculated using Equation 4 below:

$$RWP = w_1 RP + w_2 (1 - CQ) \quad \text{Equation 4}$$

where: w_1 and w_2 are weight factors of RP and CQ , which is 0.5 by default and can be adjusted through model tuning. The use of RWP for refrac can be characterized in Figure 5. Refrac wells are preferred to have RWP for higher than a set threshold based on field observations (0.5 is a good default) to balance the cost and potential performance improvements, with higher the RWP, the better the expected performance. The wells categorized with high RWP are further evaluated to see if the average production rate on those wells are on the lower side with respect to the well population to confirm as refrac candidates. For low RWP wells, it is recommended to apply lower cost restimulation options such as bullhead refracs that simply restimulate the existing fracture entry points. This is a viable option if the average production rates on these wells are on the lower end of the distribution. A cleanout or acid job treatment should be considered if a recent PI deterioration is observed to clean up near wellbore. The goal is simply restoring PI back to original trend from near wellbore damage such as sand plugging, as can be seen in a well example in Figure 5, that saw a PI drop recently which post a cleanout led to restored PI.



(a) RWP for refrac and restimulation wells



(b) Additional filter for restimulation wells

Figure 5—Refracture and restimulation well selection workflow and example

Post-refracturing Impact Analysis

The post-refrac impact analysis is designed to quantify the changes in well and reservoir performance following a refrac treatment. The analysis integrates multi-segment diagnostic workflows with production forecasting, allowing for a detailed evaluation of reservoir potential, completion performance, and productivity gains. The workflow consists of the following steps in Figure 6:

1. Identification of refracturing events using MS-DDV
2. Pre-refrac production, DDV and PI calculations
3. Post-refrac production, DDV and PI calculations
4. Quantification of incremental changes pre vs post refrac: delta DDV, delta PI, delta rate
5. Five-year production forecast and delta cumulative liquid volume
6. Post-refrac performance ranking and comparison against pre-refrac RWP ranking

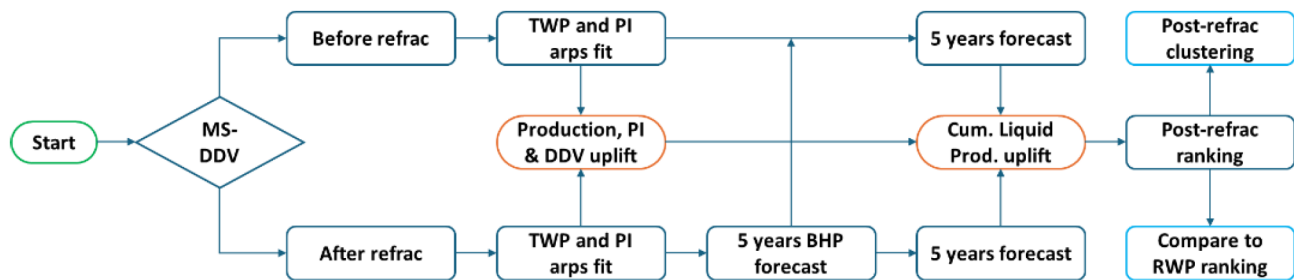


Figure 6—Post-refrac analysis flow chart

The MS-DDV workflow is applied to detect significant discontinuities or step changes in drainage volume, which typically indicate refracturing or frachit events. The appropriate category is validated using completion data. The corresponding time stamps are used as reference points for segmenting the production history into pre-refrac and post-refrac intervals. This step ensures that subsequent analyses focus on the impact of the refrac event itself rather than unrelated operational changes. Then, for the pre-refrac interval, the PI decline trend is determined by using historical production and PI data. Productivity Index (PI) and drainage volume (DDV) pre-refrac provides a baseline for subsequent performance comparison against post-refrac. Following the refrac, new DDV and PI trends are calculated using the post-refrac production history. The PI decline trend is recalibrated post the refrac event with Arps hyperbolic decline. Finally, using a common BHP forecast, a five-year production forecast can be generated using both pre-refrac and post-refrac PI decline trend analysis.

Post-refracturing performance ranking can be carried out using simple metrics such as forecasted cumulative production uplift, or through unsupervised clustering approaches that incorporate additional

quantitative parameters such as DDV, PI, and rates. For clarity and ease of comparison, this study evaluates RWP directly against forecasted cumulative production uplift to assess its effectiveness against the candidate-ranking metric. The results show a strong correlation between wells with higher RWP scores and those achieving significant post-refrac production gains, thereby validating the proposed candidate selection methodology. A separate section provides a detailed clustering analysis performed using differences in DDV, cumulative liquid production, PI, and normalized average PI forecast.

Result: Case Study

The proposed methodology was applied to a pilot study involving ~190 unconventional wells in a prominent US onshore basin with roughly 20 wells refrac'd in the past and the remaining were candidates evaluated for refrac program. The proposed refrac well ranking workflow was validated by comparing pre-job RWP rankings with post-job performance metrics, including production uplift, changes in drainage volume, and productivity index.

Post-refrac Performance Analysis

This section presents the application of the multi-segmented dynamic drainage volume (MS-DDV) methodology to identify refracturing events and evaluate post-job performance uplift. The analysis quantifies changes in reservoir/well diagnostic parameters and forecasts incremental production, which are subsequently used for validating the Refrac Well Potential (RWP) ranking.

Multi-Segmented DDV Analysis. Application of the multi-segmented DDV (MS-DDV) approach successfully identified refrac and frac-hit events, allowing for separate modeling of pre- and post-event production behavior. This improved the fidelity of reservoir pressure and productivity index estimates. As introduced in Section 2, the Sparse Identification of Nonlinear Dynamics (SINDy) method was applied to the rate-normalized pressure (RNP) versus material balance time (MBT) relationship for each production segment. This approach enables robust flow regime classification, allowing the detection of subtle transitions such as sub-linear to pseudo-steady-state behavior, to characterize dynamic drainage volume and subsequently dynamic average reservoir pressure estimates.

Figure 7 summarizes the changes in average reservoir pressure, PI, and DDV for around 20 recently refrac'd wells as evident from the discontinuities in the profiles. Overall, the majority of wells exhibited a clear uplift in all three parameters following refracturing, indicating successful stimulation and improved reservoir connectivity. However, a small subset of wells showed slight declines in one or more parameters. These negative responses are likely related to complex parent-child well interactions, which may have limited the effectiveness of the refracturing treatment. This behavior could explain the relatively poor post-refrac production performance observed for these wells in the subsequent analysis.

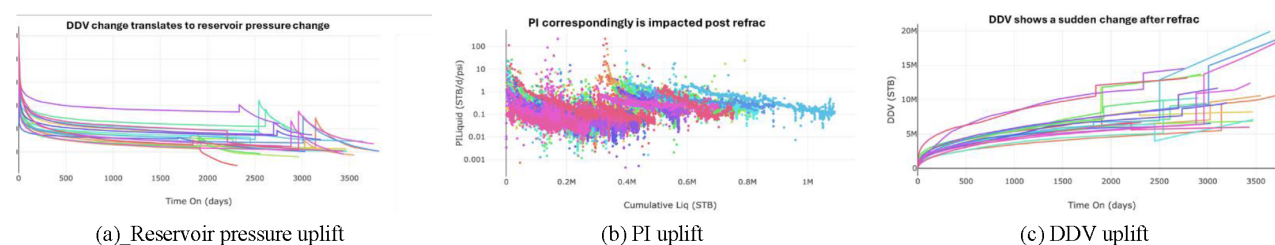


Figure 7—Reservoir pressure, PI, DDV uplift after refrac

Post-refrac Cumulative Liquid Production Uplift. As introduced in section 2.4, the post-refrac performance was evaluated by applying the PIBF workflow over 5-year post refrac with BHP forecasted for individual wells as shown in Figure 8. The idea is to compute the expected volumes produced by each refrac well post the refrac job over a duration of 5 years to quantify better performing wells in terms of

overall production. The Arps decline parameters for the post-refrac segment were obtained using only the refrac'd production history. BHP, PI, and cumulative liquid forecast after post-refrac are shown in Figure 8.

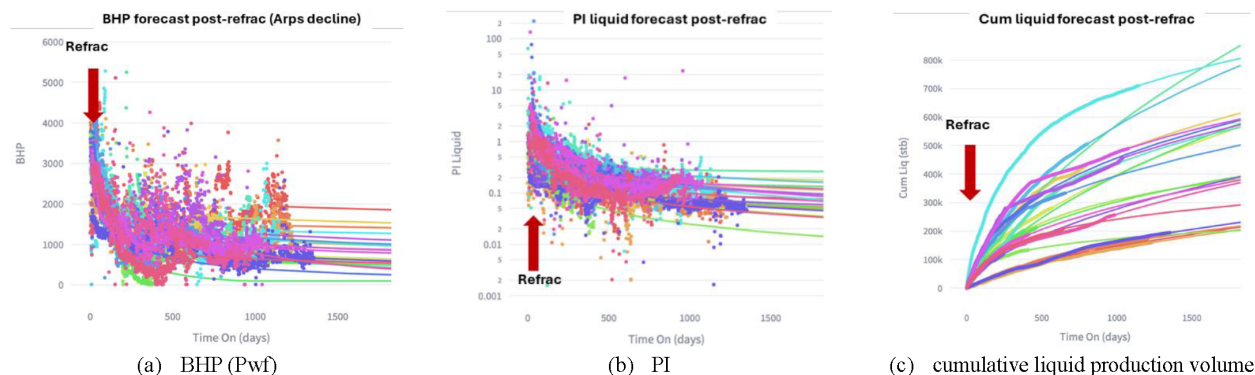


Figure 8—Post-refrac forecast for all wells using individual BHP forecasts

However, in order to compute the impact due to refrac and evaluate against the proposed ranking criteria, it is essential to compare the expected base profile against the profile post refrac. Figure 9 represents two representative examples that clearly quantify the incremental production uplift achieved following refracturing. In order to capture the true uplift achieved due to a refrac job, the pre- and post-job forecasts were run with the same BHP profile. This is possible due to the PI based forecasting workflow that allow for forecasting with any BHP profile. The five-year forecast based on the pre-refrac PI decline trend is shown by the blue line, which provides a baseline for expected production without intervention. In contrast, the red line represents the post-refrac forecast, capturing the uplift attributable to the refracturing treatment. The difference between these two forecasts quantifies the incremental production gain. These quantified uplifts provide the foundation for subsequent RWP validation analysis, in which post-job production improvements are directly compared to pre-job RWP rankings. This comparison enables assessment of the predictive accuracy and practical applicability of the proposed candidate selection methodology.

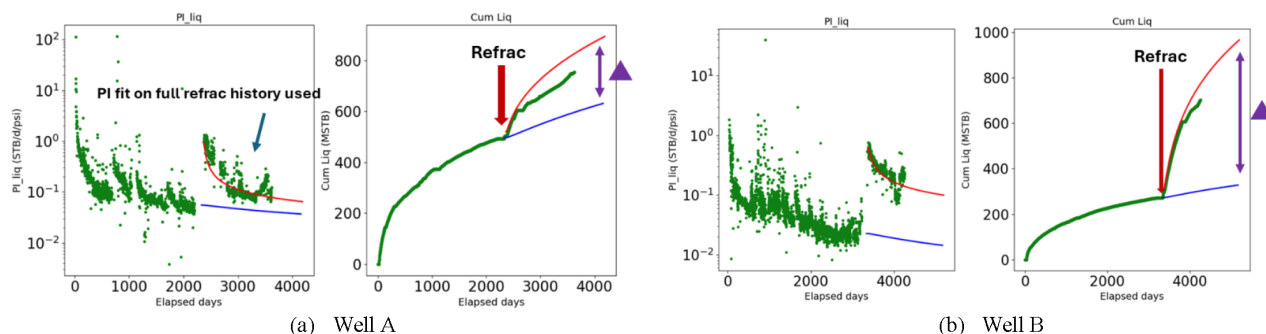


Figure 9—Post-refrac production uplift and performance analysis compared to pre-refrac forecast

Post-refrac Well Clustering. Table 3 presents heatmap from a hierarchical clustering algorithm of post-refracturing well performance using cumulative liquid uplift, DDV uplift, PI uplift, and average PI normalized by liquid volumes produced as evaluation metrics. The wells were grouped into five clusters (0–4), which delineate clear performance tiers. Cluster 4 wells (B-B-8, E-P-5, and E-U-4) represent the highest performers, exhibiting strong uplift across all metrics and validating the effectiveness of refracturing in these candidates. Cluster 3 wells also achieved substantial DDV and cumulative liquid improvements but showed more modest PI uplift, suggesting production gains were primarily driven by drainage expansion rather than significant efficiency improvements. In contrast, Clusters 1 and 2 displayed only moderate benefits, with some wells recording inconsistent or even negative DDV responses, indicating less robust outcomes,

however they had significant PI uplift contributing to increased production. Cluster 0 contained the poorest performers, dominated by bullhead refracs, which consistently showed limited uplift as expected since they are a lower cost option acting towards re-energizing only the existing fractures.

Table 3—Post-refrac well clustering results

Well Name	Cluster	Cum. Liq. Uplift	DDV uplift	PI uplift	Avg PI (normalized)	Comments
B-B-8	4	493796.73	3684502.7	1.33	0.53	Best overall
E-P-5	4	714549.87	6019163.5	1.55	0.71	
E-U-4	4	545605.07	2594673.19	1.47	0.56	
E-H-4	3	562703.49	4538813	0.35	0.35	Good examples: Higher DDV and Cum. Liq. Uplift, but lower PI uplift
E-U-2	3	747068.95	5627525.23	0.74	0.36	
E-M-3	3	792738.66	4767070.07	0.3	0.31	
E-M-2	3	496475.85	1925578.27	0.32	0.19	
E-H-3	3	638860.32	7264468.46	0.52	0.28	
B-B-6	2	514609.35	4783721.52	0.89	0.44	
E-U-3	2	500183.16	3928849.05	0.69	0.53	
B-B-12	1	263075.38	1570203.02	0.89	0.27	Med performers: fair uplift
B-B-4	1	378705.36	-1239932.47	0.55	0.45	
B-B-10	1	188666.15	1745659.99	0.85	0.3	
E-J-9	1	334030.11	1011147.47	0.63	0.24	
E-P-4	1	511355.17	-2448612.24	0.82	0.48	
E-F-5	0	165289.98	-4209.14	0.2	0.12	Poor performers: (with 4 bullhead wells)
E-F-4	0	186536.94	-660639.35	0.21	0.13	
E-F-6	0	111671.96	-1078028.02	0.14	0.1	
E-J-2	0	281337.3	2365707.4	0.35	0.19	
E-J-7	0	121696.67	2059803.43	0.38	0.18	
H-T-5	0	171932.94	-35246.89	0.12	0.09	

These results demonstrate that multivariate clustering provides a systematic means of distinguishing refracturing outcomes and identifying drivers of success or failure. Importantly, the analysis reveals strong alignment between high RWP scores and membership in the top-performing clusters, thereby reinforcing the validity of the proposed candidate ranking approach. At the same time, the clear separation of poor-performing bullhead treatments emphasizes the need for careful candidate selection and tailored operational strategies to maximize refracturing value.

Refrac Candidate Ranking Validation with Post-Refrac Performance

This section evaluates the effectiveness of the Refrac Well Potential (RWP) methodology by comparing pre-job candidate rankings with actual post-refrac performance outcomes. The objective is to determine whether wells with higher RWP scores consistently deliver greater production uplift and improved reservoir/well diagnostic parameters after refracturing. An overall validation was firstly conducted by comparing RWP ranking and post-refrac cumulative production uplift across all refrac'd wells. Then a detailed analysis was conducted for well groups by location comparing refrac'd wells against their neighboring wells in the same area.

Overall Validation with Refrac Completed Wells. The 21 refrac'd wells can be further divided groups by formation (A and B), and refrac type (general with new fracs created and Bullhead). Figure 10 presents the relationship between RWP and incremental cumulative liquid volume across these groupings. It was ensured that the refrac completion design between all these wells were not very different to enable evaluating the post refrac performance to the ranking criteria.

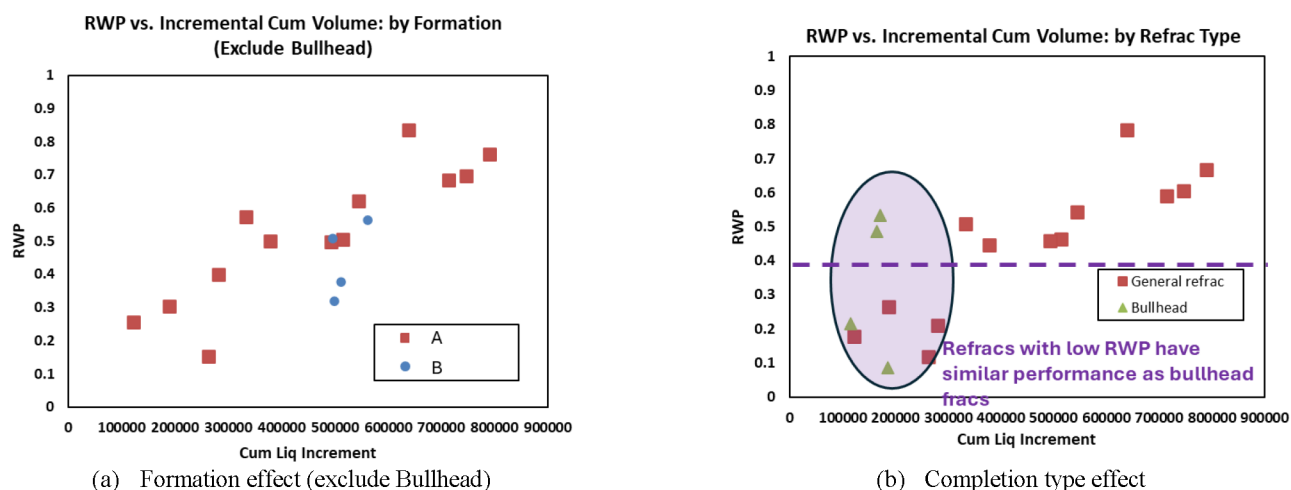


Figure 10—RWP vs. cumulative liquid production uplift

As can be seen, although wells were spread out in the field, they follow a very good linear relationship between RWP and Cum. Liq. uplift, indicating that the ranking criteria effectively maintains predictive consistency across different locations, thus validating the generalizability of parameters used in the RP and CQ definitions. Figure 10 (a) compares refrac jobs of different formation. As seen, most refrac wells in the dataset are from formation A, and the RWP methodology has been primarily tuned for this reservoir type, which shows a more linear relationship between RWP and Cum. Liq. Uplift. In contrast, formation B shows greater scatter, largely due to the limited number of refrac wells available for analysis; additional data would be beneficial to further refine the method for this formation. It should be noted that the exponents in the RP and CQ formulations were tuned to linearize this relationship as much as possible. Without tuning the exponents, it maintained a reasonably linear trend thus confirming the criteria can be used if there are no refrac wells to tune against. Tuning the exponents slightly improved the Pearson correlation to 0.85 from 0.75 before tuning, thus the model can be retuned as additional data is obtained. Figure 10 (b) illustrates the completion type effect. Wells completed with the bullhead refrac method consistently exhibit poorer performance compared to general refracs. It should be noted that bullhead refracturing involves pumping stimulation fluids directly into existing fracs from the surface without zonal isolation, which typically limits fracture initiation control and reservoir coverage. It is more akin to a stimulation treatment rather than a full refracturing operation. As can be observed, for wells with $RWP > 0.4$, general refracturing is recommended to maximize potential gains, whereas for wells with $RWP < 0.4$, lower-cost options such as bullhead treatments or other stimulation methods may be more cost-effective as regular refracs for those wells did not yield significant production gains. Overall, the results confirm that RWP maintains predictive capability across different well locations, performs best in formations with sufficient tuning data (A), and aligns with operational decision-making when considering completion type and cost-effectiveness.

RWP Ranking in Neighboring Wells: Case Study 1. As described earlier, 21 refrac'd wells were selected from multiple locations for detailed performance evaluation. In this example, 2 refrac'd wells and 17 neighboring wells were analyzed to examine pre-refrac and post-refrac performance within a local well group. Table 4 summarizes the RWP and key diagnostic parameters for the five highest-ranked wells in this group.

Table 4—RWP and corresponding parameters for case study 1

Well	Rank	RP	CQ	RWP	spacing	DDV	pavg	depletion	Days to transitional	Fraction of DTT	Formation
M3 (refrac'd)	1	0.651	0.316	0.6679	878.0942	5100300	3861.5	0.50805	[1825, 0]	0.71879	A
M4	2	0.403	0.209	0.59704	744.1866	11229462	4519.7	0.37743	[2190, 0]	0.84818	B
M2 (refrac'd)	3	0.293	0.433	0.42972	581.7452	7152036	4091.1	0.47872	[2555, 0]	0.97893	B
M6	4	0.091	1	0.04584	576.5917	9627057	4380.2	0.32102	[365]	0.82393	A
T11	5	0.075	1	0.03791	703.5888	17136667	2999.2	0.58156	[365]	0.22657	B

RWP scores of the top 3 wells in this 19-well group are 0.668, 0.597, and 0.430, respectively, of which 2 were refrac'd. These values suggest strong reservoir potential and/or suboptimal completion quality prior to intervention, making them favorable refracturing candidates. This is an example where the well M4 bounded by M2 and M3 indirectly benefited from the refrac wells due to spatial proximity possibly due to enhanced reservoir connectivity and pressure support. As shown in Figure 11, all three wells show significant improvements in DDV, cumulative PI, reservoir pressure, and production. Wells 1 and 3 exhibiting the largest DDV gains due to direct refracturing. Well 2, although not directly refrac'd, also demonstrates a measurable uplift in PI and production, albeit with a slightly smaller DDV uplift compared to the directly treated wells.

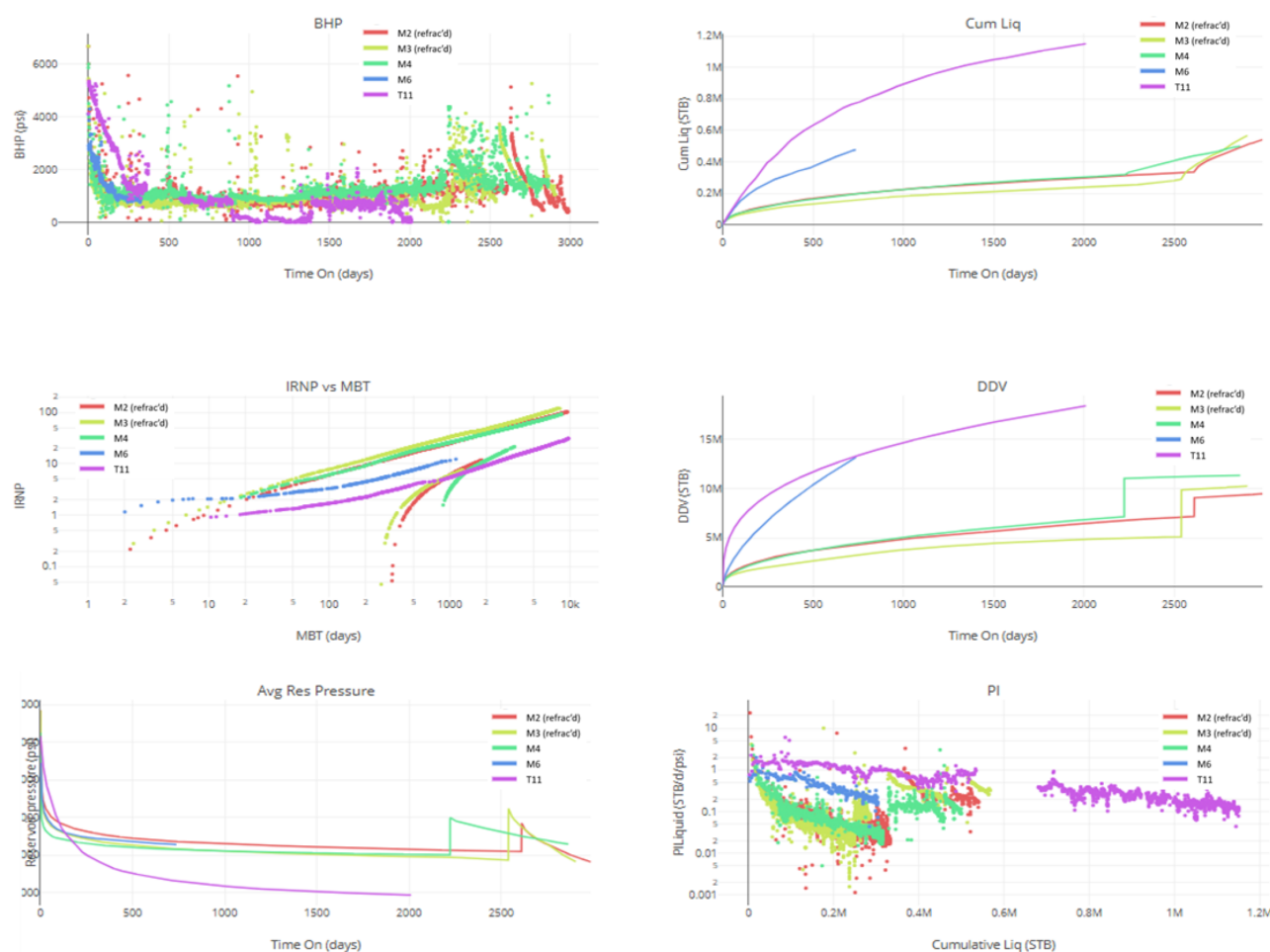


Figure 11—Reservoir performance analysis for RWP ranking case study 1

Overall, this example highlights the value of RWP-based candidate selection in identifying wells with strong potential for refracturing and validates the operators decision on choosing appropriate candidates in

this area. The observed alignment between high RWP rankings and favorable post-job performance further validates the predictive capability of the proposed methodology.

RWP Ranking in Neighboring Wells: Case Study 2. A similar example group is shown in Figure 12 and Table 5, comparing the refrac'd wells with around 35 neighboring wells in the area. This comparison shows the refrac'd wells also ranked higher among the other wells except for one that ranked slightly lower in the list. The same well did not show very good performance compared to the 2 higher ranked refracs thus confirming to the observations based on the ranked list. This list also suggests the E-11 and E-5 could have been better choices in the area for the operator owing to poor completion quality indicated in the CQ column. Thus, the analysis shows promise in quick and physically consistent way of ranking the wells for refrac without any manual assumption and evaluating the refrac program across the entire field.

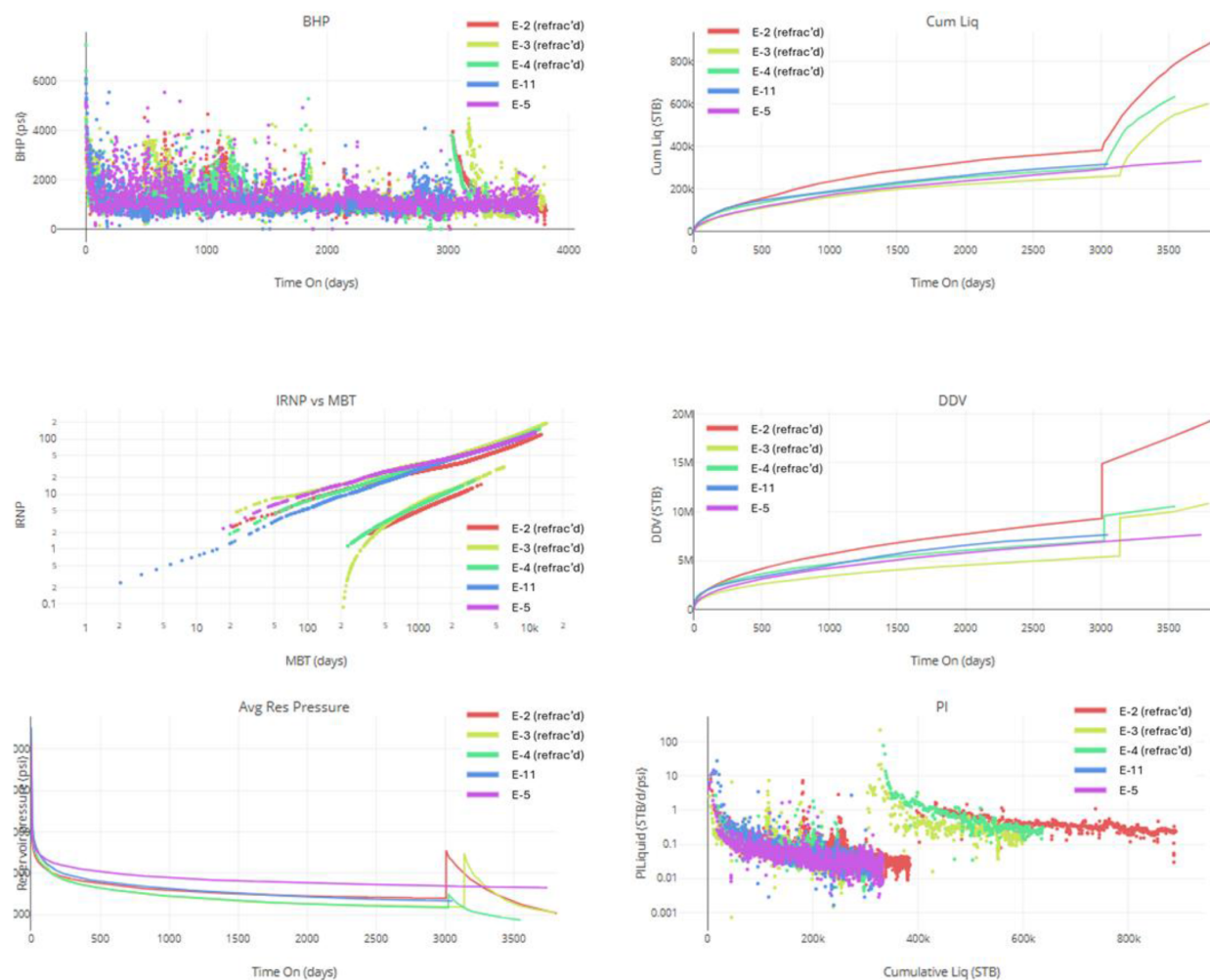


Figure 12—Reservoir performance analysis for RWP ranking case study 2

Table 5—RWP and corresponding parameters for case study 2

Well	Rank	RP	CQ	RWP	spacing	DDV	pavg	depletion	Days to transitional	Fraction of DTT	formation
E-11	1	0.21	0.00	0.61	528.6449	6845219	3405.66	0.54724	[2190]	0.99863	B
E-2 (refrac'd)	2	0.55	0.34	0.61	866.6803	9309486	3390.57	0.52406	[1825, 0]	0.60651	A
E-5	3	0.31	0.18	0.56	523.5017	6871170	3698.982	0.48222	[3285]	1	A
E-4 (refrac'd)	4	0.45	0.36	0.54	698.3956	7055704	3165.242	0.56001	[1825, 0]	0.60331	A
E-6	5	0.20	0.22	0.49	496.8732	4545121	3185.823	0.59102	[1825]	0.61283	B
E-7	6	0.12	0.20	0.46	457.493	7480413	3275.645	0.54323	[2555]	1	B
E-2	7	0.19	0.47	0.36	500.6794	4217545	3245.254	0.56888	[1825]	0.59758	B
E-9	8	0.09	0.47	0.31	365.0543	7038401	3191.835	0.57934	[1460]	0.64545	B
E-10	9	0.11	0.57	0.27	376.3779	6893585	3288.664	0.54682	[1095]	0.52493	A
E-H-5	10	0.20	0.68	0.26	449.9721	4204751	3132.191	0.59693	[1095]	0.35937	A
E-3 (refrac'd)	11	0.28	0.78	0.25	642.5851	5489180	3171.964	0.55501	[1460, 0]	0.46497	B

Well Spacing Effect. Well spacing is an important consideration by operators to pick refrac candidates with preference for larger spaced wells to avoid risk of frachits and potential to tap into undrained reservoir. This is an example that validates the workflow for two wells in the area that have very similar properties and production profiles as can be seen in Figure 13. However, the spacing on Well#1 is higher compared to that of Well#2 that causes the RP to be on the higher side for Well#1 and thus being ranked higher on RWP. The completion quality for these wells is very similar as evidenced from the PI and Pavg profiles. This example illustrates how the approach conforms to the operator preference to have higher spacing between wells for refracs but also provides more confidence by tracking other useful parameters that are not typically easy to obtain such as DDV, average reservoir pressure and PI.

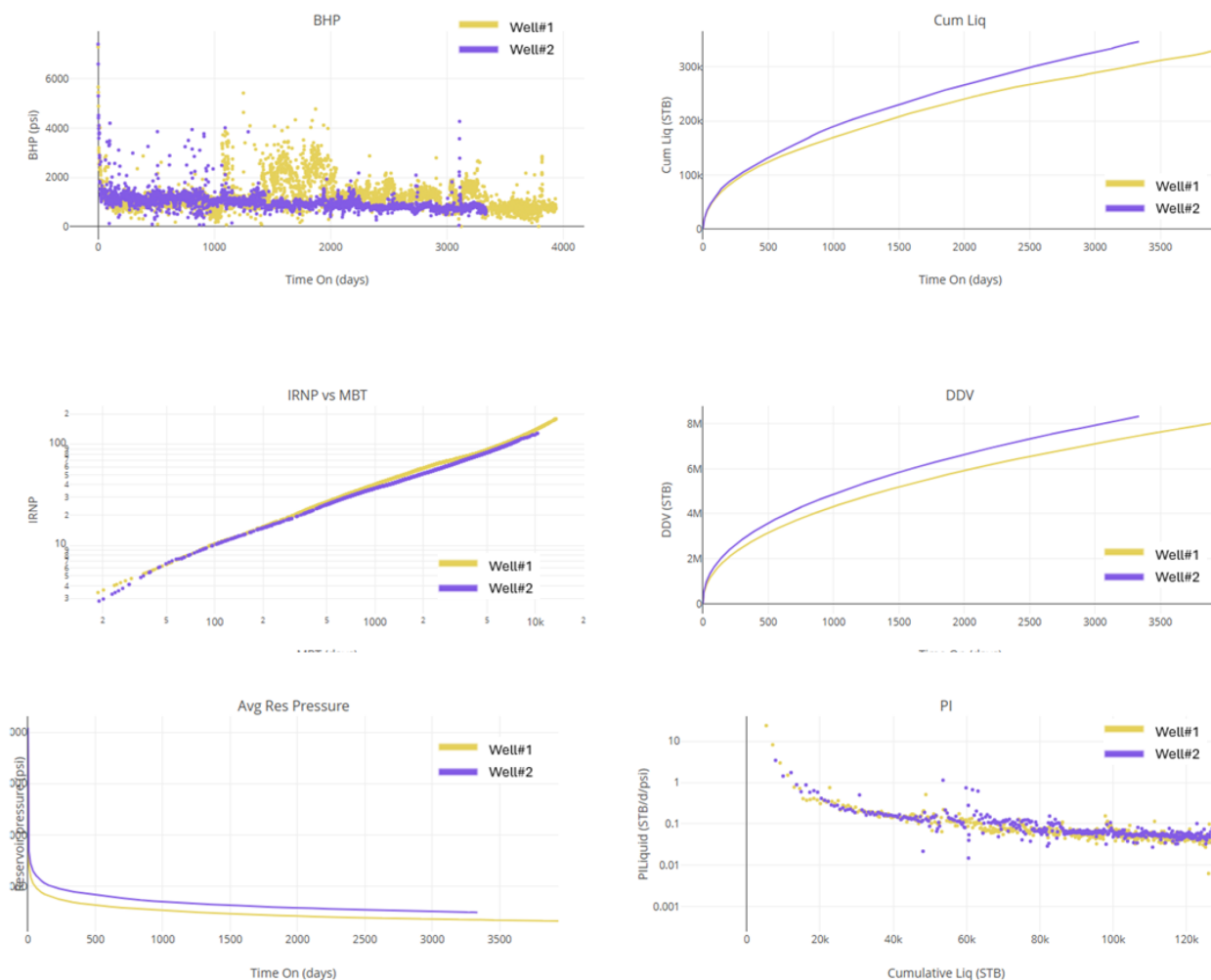


Figure 13—Reservoir performance analysis for well spacing effect

Table 6—RWP and corresponding parameters for well spacing effect

Well	Rank	RP	CQ	RWP	spacing	DDV	depletion	Days to transitional	Fraction of DTT	formation
Well#1	6	1	0.2	0.9	1184.053	8061168	0.47	3285	0.83	A
Well#2	34	0.54	0.23	0.66	782.0374	8296094	0.45	3285	0.99	A

Future Refrac Well Group Selection Case Study. Given the high cost of refracturing operations and rig constraints, it is often more practical to target an entire pad of wells rather than treating a single well in isolation. Therefore, in addition to ranking individual wells by Refrac Well Potential (RWP), it is important to evaluate pads based on average RWP to optimize both operational efficiency and economic return.

Table 7 compares two example pads—BWN with an average RWP of 0.72 and BWS with an average RWP of 0.79. Although BWN1 is the top-performing individual candidate, the BWS pad exhibits more balanced high-RWP wells, making it a stronger overall refracturing target. In addition, BWS wells generally have competitive dynamic drainage volumes (DDV), reservoir pressures, and fraction_ttf values, indicating strong reservoir potential across the pad. Treating multiple high-RWP wells within a pad (as in BWS) is likely to yield higher incremental production per unit cost compared to treating a single standout well in a lower-average-RWP pad (as in BWN). Therefore, although BWN1 ranks highest among all individual wells (RWP = 0.91), the overall pad analysis suggests that BWS offers greater aggregate refracturing potential due to higher average RWP and a more consistent distribution of well performance within the pad. This case study highlights the importance of integrating pad-level RWP evaluation into refracturing planning,

enabling operators to identify not only the best-performing wells but also the most promising groups of wells for cost-effective field development.

Table 7—RWP and corresponding parameters for future pad refrac candidate selection

Well	Rank	RP	CQ	RWP	spacing	DDV	pavg	depletion	days to transitional	fraction ttf	formation
BWN1	4	0.94	0.13	0.91	1065	7109989	4005.6	0.51	[3650]	0.95	A
BWN2	41	0.6	0.32	0.64	868	5718571	3837.4	0.53	[2920]	0.89	A
BWN3	46	0.65	0.43	0.61	922	6517315	3889.2	0.52	[3650]	1	A
BWS2	23	0.87	0.46	0.7	1013	7729533	4120.9	0.5	[3650]	0.94	A
BWS3	8	1	0.25	0.88	1134	7832362	4063.2	0.51	[3285]	0.9	A
BWS4	14	0.82	0.23	0.79	1012	6205093	4358.0	0.46	[2920]	0.89	A

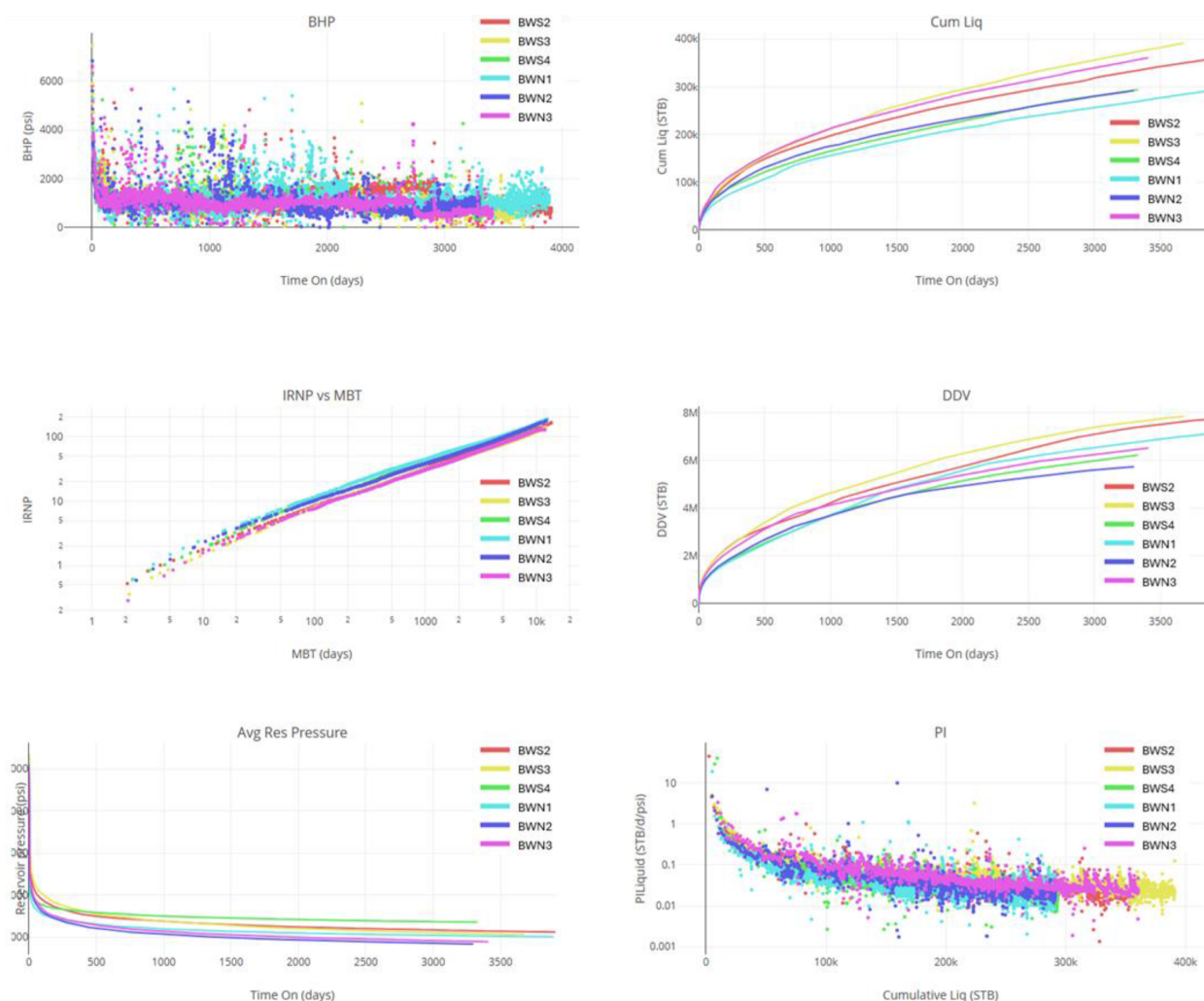


Figure 14—Reservoir performance analysis for future refrac group study

Discussion and Conclusion

This study developed and validated a novel physics-inspired, data-driven workflow that integrates MS-DDV derived reservoir diagnostics with the Refrac Well Potential (RWP) metric to improve refrac candidate selection in unconventional wells. The approach combines robust physics-based pressure and productivity diagnostics with a quantitative ranking system, enabling operators to identify wells with the highest likelihood of achieving sustained production uplift. Validation against field data confirmed the diagnostic capability of MS-DDV with SINDy-based flow regime detection, and the predictive value of RWP. Key findings are:

1. The new MS-DDV approach can successfully detect refrac and frac-hit events, enabling precise segmentation and computation of DDV and average reservoir pressures for pre- and post-event analysis.
2. Refrac well potential metric is coined that is defined based on "Reservoir Potential" and "Completion Quality" metrics using dynamic and static reservoir and completion properties.
3. The proposed RWP value can be used as an indicator for refrac candidate selection across entire field. RWP showed strong alignment with post-job production uplift on example case studies. In general, a preferred refrac well should have a higher RWP. Formation-specific calibration is recommended for broader applicability.
4. It is recommended to conduct general refracturing job for high RWP wells due to its high payback. Bullhead refrac job can be considered for low-RWP wells while restimulation should be preferred for abnormal recent PI declines observed that are a symptom of near wellbore damage.
5. Evaluating RWP at the pad or group level enables optimizing resource allocation, and improves overall project economics for a batch refrac program.

The integration of physics-based reservoir diagnostics and data-driven ranking provides a flexible, scalable toolset for unconventional reservoir operators. While undertaking future work on expanding the dataset to other unconventional plays, recalibration of the RWP formula for diverse geological settings need to be considered that is possible through tunable exponents introduced in the formulation.

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